

MathCity.org**Merging man and maths****NOTES OF EXERCISE 9.3****Version:1.0****❖ Homogenous functions:****➤ Definition:**

A function $f(x, y)$ is called homogenous of degree "n" if

$$f(tx, ty) = t^n f(x, y)$$

Where t is a nonzero real number.

❖ Homogenous functions:**➤ Definition:**

A first order differential equation

$$\frac{dy}{dx} = f(x, y)$$

is said to be homogenous if f is a homogenous function of any degree.

Example:

$\frac{dy}{dx} = \frac{x+y}{x-y}$ is an example of homogenous equation.

❖ Theorem:

A homogenous equation $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$ can be transformed into a separable equation by the substitution $y = vx$.

❖ EXERCISE 9.3

Question #1: $(x - y)dx + (x + y)dy = 0$ **Solution:**

Given equation is

$$(x - y)dx + (x + y)dy = 0$$

$$\Rightarrow (x + y)dy = -(x - y)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{-(x - y)}{(x + y)} \quad \dots (a)$$

This is a homogenous differential equation in x & y . To solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{-(x - vx)}{(x + vx)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{-x(1 - v)}{x(1 + v)}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v - 1}{v + 1} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v - 1 - v^2 - v}{v + 1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{-(1 + v^2)}{v + 1}$$

$$\Rightarrow \frac{v + 1}{1 + v^2} dv = -\frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{v + 1}{1 + v^2} dv = - \int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \int \frac{2v}{1 + v^2} dv + \int \frac{dv}{1 + v^2} = - \int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \ln(1 + v^2) + \tan^{-1} v = -\ln x + \ln c$$

$$\Rightarrow \frac{1}{2} \ln \left(1 + \frac{y^2}{x^2} \right) + \tan^{-1} \frac{y}{x} = -\ln x + c$$

$$\Rightarrow \frac{1}{2} \ln \left(\frac{x^2 + y^2}{x^2} \right) + \tan^{-1} \frac{y}{x} = -\ln x + c$$

$$\Rightarrow \frac{1}{2} \ln(x^2 + y^2) - \frac{1}{2} \ln(x^2) + \tan^{-1} \frac{y}{x} = -\ln x + c$$

$$\Rightarrow \frac{1}{2} \ln(x^2 + y^2) - \ln(x) + \tan^{-1} \frac{y}{x} = -\ln x + \ln c$$

$$\Rightarrow \frac{1}{2} \ln(x^2 + y^2) + \tan^{-1} \frac{y}{x} = \ln c$$

is required solution.

❖ **Question #2:** $(y^2 + 2xy)dx + x^2 dy = 0$ **Solution:**

Given equation is

$$(y^2 + 2xy)dx + x^2 dy = 0$$

$$\Rightarrow \frac{dy}{dx} = -y^2 \sin x$$

$$\Rightarrow x^2 dy = -(y^2 + 2xy)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{-(y^2 + 2xy)}{x^2} \quad \text{--- (a)}$$

This is a homogenous differential equation in x & y . To solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{-(v^2 x^2 + 2x^2 v)}{x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{-x^2(v^2 + 2v)}{x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = -v^2 - 2v$$

$$\Rightarrow x \frac{dv}{dx} = -v^2 - 2v - v$$

$$\Rightarrow x \frac{dv}{dx} = -v(v + 3)$$

$$\Rightarrow \frac{dv}{v(v + 3)} = -\frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{dv}{v(v + 3)} = -\int \frac{dx}{x} \quad \text{--- (b)}$$

Suppose that

$$\frac{1}{v(v + 3)} = \frac{A}{v} + \frac{B}{v + 3}$$

$$\Rightarrow 1 = A(v + 3) + Bv \quad \text{--- (c)}$$

Put $v = 0$ in equation (c), we get

$$1 = A(3)$$

$$\Rightarrow A = \frac{1}{3}$$

Put $v + 3 = 0 \Rightarrow v = -3$ in equation (c), we get

$$1 = B(-3)$$

$$\Rightarrow B = -\frac{1}{3}$$

Hence,

$$\frac{1}{v(v + 3)} = \frac{1}{3v} - \frac{1}{3(v + 3)}$$

Thus, equation (b) becomes

$$\int \left[\frac{1}{3v} - \frac{1}{3(v + 3)} \right] dv = -\int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{3} \int \frac{dv}{v} - \frac{1}{3} \int \frac{dv}{v + 3} = -\ln x + \ln c$$

$$\Rightarrow \frac{1}{3} \ln v - \frac{1}{3} \ln(v + 3) = \ln \frac{c}{x}$$

$$\Rightarrow \frac{1}{3} [\ln v - \ln(v + 3)] = \ln \frac{c}{x}$$

$$\Rightarrow \ln \frac{v}{v + 3} = 3 \ln \frac{c}{x}$$

$$\Rightarrow \ln \frac{v}{v + 3} = \ln \frac{c^3}{x^3}$$

$$\Rightarrow \ln \frac{(y/x)}{(y/x + 3)} = \ln \frac{c^3}{x^3} \because v = \frac{y}{x}$$

$$\Rightarrow \ln \left(\frac{y}{y + 3x} \right) = \ln \frac{c^3}{x^3} \because c^3 = c \text{ (a constant)}$$

$$\Rightarrow \frac{y}{y + 3x} = \frac{c}{x^3}$$

$$\Rightarrow y|x^3| = c|y + 3x|$$

is required solution.

❖ **Question # 3:**

$$(x^2 - 3y^2)dx + 2xydy = 0$$

Solution:

Given equation is

$$(x^2 - 3y^2)dx + 2xydy = 0$$

$$\Rightarrow 2xydy = (3y^2 - x^2)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{3y^2 - x^2}{2xy} \text{ --- (a)}$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{3v^2x^2 - x^2}{2vx^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^2(3v^2 - 1)}{2vx^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{3v^2 - 1}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{3v^2 - 1}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{3v^2 - 1 - 2v^2}{2v} = \frac{v^2 - 1}{2v}$$

$$\Rightarrow \frac{2v}{v^2 - 1} dv = \frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{2v}{v^2 - 1} dv = \int \frac{dx}{x}$$

$$\Rightarrow \ln(v^2 - 1) = \ln x + \ln c$$

$$\Rightarrow \ln(v^2 - 1) = \ln cx$$

$$\Rightarrow v^2 - 1 = cx$$

$$\Rightarrow \frac{y^2}{x^2} - 1 = cx \quad \because v = \frac{y}{x}$$

$$\Rightarrow |y^2 - x^2| = |cx|x^2$$

is required solution.

❖ **Question # 4:**

$$3x\cos\left(\frac{y}{x}\right)dy = \left[2x\sin\left(\frac{y}{x}\right) + 3y\cos\left(\frac{y}{x}\right)\right]dy$$

Solution:

Given equation is

$$3x\cos\left(\frac{y}{x}\right)dy = \left[2x\sin\left(\frac{y}{x}\right) + 3y\cos\left(\frac{y}{x}\right)\right]dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x\sin\left(\frac{y}{x}\right) + 3y\cos\left(\frac{y}{x}\right)}{3x\cos\left(\frac{y}{x}\right)} \text{ --- (a)}$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{2x\sin v + 3vx\cos v}{3x\cos v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2x\sin v + 3vx\cos v}{3x\cos v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2x\sin v}{3x\cos v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2}{3} \tan v$$

$$\Rightarrow \frac{dv}{\tan v} = \frac{2}{3} \frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{dv}{\tan v} = \frac{2}{3} \int \frac{dx}{x}$$

$$\Rightarrow \int \cot v \, dv = \frac{2}{3} \int \frac{dx}{x}$$

$$\Rightarrow \ln \sin v = \frac{2}{3} \ln x + \ln c'$$

$$\Rightarrow 3 \ln \sin v = 2 \ln x + 3 \ln c'$$

$$\Rightarrow \ln \sin v^3 = \ln x^2 + \ln c'^3$$

$\because c'$ is a constant.

$$\Rightarrow c'^3 = c \text{ (we say)}$$

$$\Rightarrow \ln \sin v^3 = \ln x^2 + \ln c$$

$$\Rightarrow \ln \sin v^3 = \ln cx^2$$

$$\Rightarrow \sin v^3 = cx^2$$

$$\Rightarrow \left| \sin \left(\frac{y}{x} \right) \right|^3 = cx^2$$

is required solution.

❖ **Question # 5:**

$$(x^2 + xy + y^2)dx - x^2dy = 0$$

Solution:

Given equation is

$$(x^2 + xy + y^2)dx - x^2dy = 0$$

$$\Rightarrow x^2dy = (x^2 + xy + y^2)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + xy + y^2}{x^2} \text{ --- (a)}$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{x^2 + vx^2 + v^2x^2}{x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^2(1 + v + v^2)}{x^2}$$

$$\Rightarrow x \frac{dv}{dx} = 1 + v + v^2 - v$$

$$\Rightarrow x \frac{dv}{dx} = 1 + v^2$$

$$\Rightarrow \frac{dv}{1 + v^2} = \frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{dv}{1 + v^2} = \int \frac{dx}{x}$$

$$\Rightarrow \tan^{-1} v = \ln x + c$$

$$\Rightarrow \tan^{-1} \left(\frac{y}{x} \right) = \ln x + c$$

$$\Rightarrow \tan^{-1} \left(\frac{y}{x} \right) - \ln|x| = c$$

is required solution.

❖ **Question # 6:**

$$ydy + xdx = \sqrt{x^2 + y^2}dx$$

Solution:

Given equation is

$$ydy + xdx = \sqrt{x^2 + y^2}dx$$

$$\Rightarrow ydy = (\sqrt{x^2 + y^2} - x) dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{x^2 + y^2} - x}{y} \quad \dots (a)$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{\sqrt{x^2 + v^2x^2} - x}{vx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{\sqrt{1 + v^2} - 1}{v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{\sqrt{1 + v^2} - 1}{v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{\sqrt{1 + v^2} - 1 - v^2}{v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{\sqrt{1 + v^2} - (1 + v^2)}{v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{\sqrt{1 + v^2}(1 - \sqrt{1 + v^2})}{v}$$

$$\Rightarrow \frac{v dv}{\sqrt{1 + v^2}(1 - \sqrt{1 + v^2})} = \frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{v dv}{\sqrt{1 + v^2}(1 - \sqrt{1 + v^2})} = \int \frac{dx}{x} \quad \dots (b)$$

Consider

$$I = \int \frac{v dv}{\sqrt{1 + v^2}(1 - \sqrt{1 + v^2})}$$

$$\text{Put } 1 - \sqrt{1 + v^2} = t$$

$$\Rightarrow -\frac{1}{2}(1 + v^2)^{-1/2} 2v dv = dt$$

$$\Rightarrow \frac{v dv}{\sqrt{1 + v^2}} = -dt$$

Therefore,

$$I = - \int \frac{dt}{t}$$

$$\Rightarrow I = -\ln t$$

$$\Rightarrow I = -\ln(1 - \sqrt{1 + v^2})$$

$$\Rightarrow \int \frac{v dv}{\sqrt{1 + v^2}(1 - \sqrt{1 + v^2})} = -\ln(1 - \sqrt{1 + v^2})$$

Hence (b) will becomes

$$-\ln(1 - \sqrt{1 + v^2}) = \ln x + \ln c'$$

$$\Rightarrow \ln(1 - \sqrt{1 + v^2}) = -\ln c'x$$

$$\Rightarrow \ln(1 - \sqrt{1 + v^2}) = \ln(c'x)^{-1}$$

$$\Rightarrow (1 - \sqrt{1 + v^2}) = (c'x)^{-1}$$

$$\Rightarrow 1 - \sqrt{1 + \frac{y^2}{x^2}} = (c'x)^{-1}$$

$$\Rightarrow 1 - \frac{\sqrt{x^2 + y^2}}{x} = \frac{1}{c'x}$$

$$\Rightarrow \frac{x - \sqrt{x^2 + y^2}}{x} = \frac{c}{x} \because \frac{1}{c'} = c$$

$$\Rightarrow x - \sqrt{x^2 + y^2} = c$$

$$\Rightarrow -\sqrt{x^2 + y^2} = c - x$$

$$\Rightarrow \left(-\sqrt{x^2 + y^2}\right)^2 = (c - x)^2$$

$$\Rightarrow x^2 + y^2 = c^2 + x^2 - 2cx$$

$$\Rightarrow y^2 = c^2 - 2cx$$

$$\Rightarrow y^2 + 2cx - c^2 = 0$$

is required solution.

❖ **Question # 7:**

$$\frac{dy}{dx} = \frac{4y - 3x}{2x - y}$$

Solution:

Given equation is

$$\frac{dy}{dx} = \frac{4y - 3x}{2x - y} \quad \dots (a)$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$\Rightarrow v + x \frac{dv}{dx} = \frac{4vx - 3x}{2x - vx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{4v - 3}{2 - v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{4v - 3}{2 - v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{4v - 3 - 2v + v^2}{2 - v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 + 2v - 3}{2 - v}$$

Integrating both sides, we have

$$\int \frac{(2 - v)}{v^2 + 2v - 3} dv = \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{(2 - v)}{(v + 3)(v - 1)} dv = \int \frac{dx}{x} \quad \dots (b)$$

Suppose

$$\frac{2 - v}{(v + 3)(v - 1)} = \frac{A}{v + 3} + \frac{B}{v - 1}$$

$$\Rightarrow 2 - v = A(v - 1) + B(v + 3) \quad \dots (i)$$

Put $v + 3 = 0 \Rightarrow v = -3$ in (i), we have

$$2 + 3 = A(-4)$$

$$\Rightarrow A = -\frac{5}{4}$$

Put $v - 1 = 0 \Rightarrow v = 1$ in (i), we have

$$2 - 1 = B(4)$$

$$\Rightarrow B = \frac{1}{4}$$

Thus equation (i) becomes

$$\frac{2 - v}{(v + 3)(v - 1)} = \frac{-5}{4(v + 3)} + \frac{1}{4(v - 1)}$$

Thus equation (b) becomes

$$\int \left(\frac{-5}{4(v + 3)} + \frac{1}{4(v - 1)} \right) dv = \int \frac{dx}{x}$$

$$\Rightarrow -\frac{5}{4} \int \frac{dv}{v + 3} + \frac{1}{4} \int \frac{dv}{v - 1} = \int \frac{dx}{x}$$

$$\Rightarrow -\frac{5}{4} \ln(v + 3) + \frac{1}{4} \ln(v - 1) = \ln x + \ln c$$

$$\Rightarrow -5 \ln(v+3) + \ln(v-1) = 4(\ln x + \ln c)$$

$$\Rightarrow \ln(v+3)^{-5} + \ln(v-1) = \ln(cx)^4$$

$$\Rightarrow \ln \frac{(v-1)}{(v+3)^5} = \ln(cx)^4$$

$$\Rightarrow \frac{(v-1)}{(v+3)^5} = (cx)^4$$

$$\Rightarrow \frac{\left(\frac{y}{x} - 1\right)}{\left(\frac{y}{x} + 3\right)^5} = (cx)^4 \quad \because v = \frac{y}{x}$$

$$\Rightarrow \frac{(y-x)x^4}{(y+3x)^5} = (cx)^4$$

$$\Rightarrow |y-x| = c|y+3x|^5$$

is required solution.

❖ **Question # 8:**

$$x \sin\left(\frac{y}{x}\right) dy = \left[y \sin\left(\frac{y}{x}\right) - x\right] dx$$

Solution:

Given equation is

$$x \sin\left(\frac{y}{x}\right) dy = \left[y \sin\left(\frac{y}{x}\right) - x\right] dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \sin\left(\frac{y}{x}\right) - x}{x \sin\left(\frac{y}{x}\right)}$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{vx \sin v - x}{x \sin v}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v \sin v - 1}{\sin v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v \sin v - 1}{\sin v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v \sin v - 1 - v \sin v}{\sin v}$$

$$\Rightarrow x \frac{dv}{dx} = -\frac{1}{\sin v}$$

$$\Rightarrow -\sin v dv = \frac{dx}{x}$$

Integrating both sides, we have

$$-\int \sin v dv = \int \frac{dx}{x}$$

$$\Rightarrow \cos v = \ln x + c$$

$$\Rightarrow \cos \frac{y}{x} = \ln x + c \quad \because v = \frac{y}{x}$$

is required solution.

❖ **Question # 9:**

$$(x^2 + y^2 \sqrt{x^2 + y^2}) dx - xy \sqrt{x^2 + y^2} dy = 0$$

Solution:

Given equation is

$$(x^2 + y^2 \sqrt{x^2 + y^2}) dx - xy \sqrt{x^2 + y^2} dy = 0$$

$$\Rightarrow xy \sqrt{x^2 + y^2} dy = (x^2 + y^2 \sqrt{x^2 + y^2}) dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + y^2 \sqrt{x^2 + y^2}}{xy \sqrt{x^2 + y^2}}$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^2 + v^2 x^2 \sqrt{x^2 + v^2 x^2}}{x(vx) \sqrt{x^2 + v^2 x^2}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^3 + v^2 x^3 \sqrt{1 + v^2}}{vx^3 \sqrt{1 + v^2}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 + v^2 \sqrt{1 + v^2}}{v \sqrt{1 + v^2}}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2 \sqrt{1 + v^2}}{v \sqrt{1 + v^2}} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2 \sqrt{1 + v^2} - v^2 \sqrt{1 + v^2}}{v \sqrt{1 + v^2}}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1}{v \sqrt{1 + v^2}}$$

$$\Rightarrow v \sqrt{1 + v^2} dv = \frac{dx}{x}$$

Integrating both sides, we have

$$\int v \sqrt{1 + v^2} dv = \int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \int 2v \sqrt{1 + v^2} dv = \ln x + \ln c$$

$$\Rightarrow \frac{1}{2} \frac{(1 + v^2)^{\frac{3}{2}}}{\frac{3}{2}} = \ln(cx)$$

$$\Rightarrow \frac{1}{3} \left(1 + \frac{y^2}{x^2}\right)^{\frac{3}{2}} = \ln(cx) \because v = \frac{y}{x}$$

$$\Rightarrow \left(\frac{x^2 + y^2}{x^2}\right)^{\frac{3}{2}} = 3 \ln(cx)$$

$$\Rightarrow \frac{(x^2 + y^2)^3}{x^3} = \ln(cx^3).$$

$$\Rightarrow (x^2 + y^2)^3 = x^3 \ln(cx^3)$$

is required solution.

❖ Question # 10:

$$(\sqrt{x+y} + \sqrt{x-y})dx - (\sqrt{x+y} - \sqrt{x-y})dy = 0$$

Solution:

Given equation is

$$(\sqrt{x+y} + \sqrt{x-y})dx - (\sqrt{x+y} - \sqrt{x-y})dy = 0$$

$$\Rightarrow (\sqrt{x+y} - \sqrt{x-y})dy = (\sqrt{x+y} + \sqrt{x-y})dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{x+y} + \sqrt{x-y}}{\sqrt{x+y} - \sqrt{x-y}} \quad \dots (a)$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{\sqrt{x+vx} + \sqrt{x-vx}}{\sqrt{x+vx} - \sqrt{x-vx}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{\sqrt{1+v} + \sqrt{1-v}}{\sqrt{1+v} - \sqrt{1-v}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{\sqrt{1+v} + \sqrt{1-v}}{\sqrt{1+v} - \sqrt{1-v}} \times \frac{\sqrt{1+v} + \sqrt{1-v}}{\sqrt{1+v} + \sqrt{1-v}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1+v+1-v+2\sqrt{1-v^2}}{1+v-1-v}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{2+2\sqrt{1-v^2}}{2v}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{2(1+\sqrt{1-v^2})}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+\sqrt{1-v^2}}{v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+\sqrt{1-v^2}-v^2}{v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{\sqrt{1-v^2} + (1-v^2)}{v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{\sqrt{1-v^2}(1+\sqrt{1-v^2})}{v}$$

$$\Rightarrow \frac{v dv}{\sqrt{1-v^2}(1+\sqrt{1-v^2})} = \frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{v dv}{\sqrt{1-v^2}(1+\sqrt{1-v^2})} = \int \frac{dx}{x}$$

$$\text{Put } 1 + \sqrt{1-v^2} = t$$

$$\Rightarrow \frac{1}{2} (1-v^2)^{-\frac{1}{2}} (-2v) dv = dt$$

$$\Rightarrow \frac{v dv}{\sqrt{1-v^2}} = -dt$$

therefore,

$$\int \frac{-dt}{t} = \int \frac{dx}{x}$$

$$\Rightarrow -\int \frac{dt}{t} = \int \frac{dx}{x}$$

$$\Rightarrow -\ln t = \ln x + \ln c$$

$$\Rightarrow -\ln(1 + \sqrt{1-v^2}) = \ln x + \ln c$$

$$\Rightarrow -\ln(1 + \sqrt{1-v^2}) = \ln cx$$

$$\Rightarrow \ln(1 + \sqrt{1-v^2}) = -\ln cx$$

$$\Rightarrow \ln(1 + \sqrt{1-v^2}) = \ln(cx)^{-1}$$

$$\Rightarrow (1 + \sqrt{1-v^2}) = \frac{1}{cx}$$

$$\Rightarrow 1 + \sqrt{1 - \frac{y^2}{x^2}} = \frac{1}{cx} \quad \because v = \frac{y}{x}$$

$$\Rightarrow 1 + \frac{\sqrt{x^2 - y^2}}{x} = \frac{1}{cx}$$

$$\Rightarrow \frac{x + \sqrt{x^2 - y^2}}{x} = \frac{1}{cx}$$

$$\Rightarrow x + \sqrt{x^2 - y^2} = \frac{1}{c} = c \text{ (a constant)}$$

$$\Rightarrow x + \sqrt{x^2 - y^2} = c$$

is required solution.

Solve the initial value problem

❖ **Question # 11:**

$$\frac{dy}{dx} = \frac{x+y}{x} \quad y(1) = 1$$

Solution:

Given equation is

$$\frac{dy}{dx} = \frac{x+y}{x} \quad \dots (a)$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{x+vx}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = 1 + v$$

$$\Rightarrow x \frac{dv}{dx} = 1 + v - v$$

$$\Rightarrow x \frac{dv}{dx} = 1 + v - v$$

$$\Rightarrow dv = \frac{dx}{x}$$

Integrating both sides, we have

$$\int dv = \int \frac{dx}{x}$$

$$\Rightarrow v = \ln x + c$$

$$\Rightarrow \frac{y}{x} = \ln x + c \quad \dots (b) \quad \because v = \frac{y}{x}$$

Applying the condition $y(1) = 1$ on (b), we have

$$1 = 0 + c$$

$$\Rightarrow c = 1$$

Therefore,

$$\frac{y}{x} = \ln x + 1$$

$$\Rightarrow y = x \ln x + x$$

is required solution.

❖ **Question # 12:**

$$(y + \sqrt{x^2 + y^2}) dx - x dy = 0 \quad y(1) = 0$$

Solution:

Given equation is

$$(y + \sqrt{x^2 + y^2}) dx - x dy = 0$$

$$\Rightarrow x dy = (y + \sqrt{x^2 + y^2}) dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} \quad \dots (a)$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{vx + \sqrt{x^2 + v^2 x^2}}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{vx + x\sqrt{1+v^2}}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = v + \sqrt{1+v^2}$$

$$\Rightarrow x \frac{dv}{dx} = v + \sqrt{1+v^2} - v$$

$$\Rightarrow x \frac{dv}{dx} = \sqrt{1+v^2}$$

$$\Rightarrow \frac{dv}{\sqrt{1+v^2}} = \frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{dv}{\sqrt{1+v^2}} = \int \frac{dx}{x}$$

$$\Rightarrow \sinh^{-1} v = \ln x + c$$

$$\Rightarrow \sinh^{-1} \left(\frac{y}{x} \right) = \ln x + c \quad \dots (b) \because v = \frac{y}{x}$$

Applying the condition $y(1) = 0$ on (b), we have

$$\sinh^{-1} \left(\frac{0}{1} \right) = \ln 1 + c$$

$$\Rightarrow c = 0$$

Therefore (b) becomes

$$\sinh^{-1} \left(\frac{y}{x} \right) = \ln x$$

$$\because \sinh^{-1} x = \ln \left(x + \sqrt{x^2 + 1} \right). \text{ Therefore,}$$

$$\Rightarrow \ln \left(\frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} \right) = \ln x$$

$$\Rightarrow \frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} = x$$

$$\Rightarrow \frac{y + \sqrt{x^2 + y^2}}{x} = x$$

$$\Rightarrow y + \sqrt{x^2 + y^2} = x^2$$

is required solution.

❖ Question # 13:

$$(2x - 5y)dx + (4x - y)dy = 0 \quad y(1) = 4$$

Solution:

Given equation is

$$(2x - 5y)dx + (4x - y)dy = 0$$

$$\Rightarrow (4x - y)dy = -(2x - 5y)dx$$

$$\Rightarrow (4x - y)dy = (5y - 2x)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{5y - 2x}{4x - y} \quad \dots (a)$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{5vx - 2x}{4x - vx}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{5v - 2}{4 - v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{5v - 2}{4 - v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{5v - 2 - 4v + v^2}{4 - v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 + v - 2}{4 - v}$$

$$\Rightarrow \frac{4 - v}{v^2 + v - 2} dv = \frac{dx}{x}$$

$$\Rightarrow \frac{4 - v}{(v + 2)(v - 1)} dv = \frac{dx}{x} \quad \dots (b)$$

Consider that

$$\frac{4-v}{(v+2)(v-1)} = \frac{A}{v+2} + \frac{B}{v-1}$$

$$\Rightarrow 4-v = A(v-1) + B(v+2) \dots (i)$$

Put $v+2=0 \Rightarrow v=-2$ in (i), we have

$$6 = A(-3)$$

$$\Rightarrow A = -2$$

Put $v-1=0 \Rightarrow v=1$ in (i), we have

$$3 = B(3)$$

$$\Rightarrow B = 1$$

therefore,

$$\frac{4-v}{(v+2)(v-1)} = \frac{-2}{v+2} + \frac{1}{v-1}$$

Thus equation (b) becomes

$$\frac{-2}{v+2} + \frac{1}{v-1} = \frac{dx}{x}$$

Integrating both sides, we have

$$\int \left(\frac{-2}{v+2} + \frac{1}{v-1} \right) dv = \int \frac{dx}{x}$$

$$-2 \int \frac{dv}{v+2} + \int \frac{dv}{v-1} = \int \frac{dx}{x}$$

$$\Rightarrow -2 \ln(v+2) + \ln(v-1) = \ln x + \ln c$$

$$\Rightarrow \ln(v+2)^{-2} + \ln(v-1) = \ln cx$$

$$\Rightarrow \ln \frac{(v-1)}{(v+2)^2} = \ln cx$$

$$\Rightarrow \frac{(v-1)}{(v+2)^2} = cx$$

$$\Rightarrow \frac{\frac{y}{x} - 1}{\left(\frac{y}{x} + 2\right)^2} = cx \quad \because v = \frac{y}{x}$$

$$\frac{(y-x)/x}{(y+2x)^2/x^2} = cx$$

$$\Rightarrow \frac{(y-x)x}{(y+2x)^2} = cx$$

$$\Rightarrow \frac{(y-x)x}{(y+2x)^2} = cx$$

$$\Rightarrow y-x = c(y+2x)^2 \dots (c)$$

Applying the condition $y(1) = 4$ on (b), we have

$$4-1 = c(4+2)^2$$

$$\Rightarrow c = \frac{1}{12}$$

$$(c) \Rightarrow y-x = \frac{1}{12}(y+2x)^2$$

$$\Rightarrow (y+2x)^2 = 12(y-x)$$

is required solution.

❖ Question # 14:

$$(3x^2 + 9xy + 5y^2)dx - (6x^2 + 4xy)dy = 0 \quad y(2) = -6$$

Solution:

Given equation is

$$(3x^2 + 9xy + 5y^2)dx - (6x^2 + 4xy)dy = 0$$

$$\Rightarrow (6x^2 + 4xy)dy = (3x^2 + 9xy + 5y^2)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{(3x^2 + 9xy + 5y^2)}{(6x^2 + 4xy)} \dots (a)$$

This is a homogenous differential equation in x & y . to solve this, put

$$y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus equation (a) becomes

$$v + x \frac{dv}{dx} = \frac{3x^2 + 9vx^2 + 5v^2x^2}{6x^2 + 4vx^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{3 + 9v + 5v^2}{6 + 4v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{3 + 9v + 5v^2}{6 + 4v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{3 + 9v + 5v^2 - 6v - 4v^2}{6 + 4v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 + 3v + 3}{6 + 4v}$$

$$\Rightarrow \frac{6 + 4v}{v^2 + 3v + 3} dv = \frac{dx}{x}$$

Integrating both sides, we have

$$\int \frac{6 + 4v}{v^2 + 3v + 3} dv = \int \frac{dx}{x}$$

$$\Rightarrow 2 \int \frac{(2v + 3)}{v^2 + 3v + 3} dv = \int \frac{dx}{x}$$

$$\Rightarrow 2 \ln(v^2 + 3v + 3) = \ln x + \ln c$$

$$\Rightarrow \ln(v^2 + 3v + 3)^2 = \ln cx$$

$$\Rightarrow (v^2 + 3v + 3)^2 = cx$$

$$\Rightarrow \left(\frac{y^2}{x^2} + 3 \frac{y}{x} + 3 \right)^2 = cx$$

$$\Rightarrow \left(\frac{y^2 + 3xy + 3x^2}{x^2} \right)^2 = cx$$

$$\Rightarrow (y^2 + 3xy + 3x^2)^2 = cx^5 \text{ --- (b)}$$

Applying the condition $y(2) = 6$ on (b), we have

$$(36 - 36 + 12)^2 = c(8)$$

$$\Rightarrow 144 = 32c$$

$$\Rightarrow \frac{144}{32} = c$$

$$\Rightarrow c = \frac{9}{2}$$

Therefore (b) becomes

$$(y^2 + 3xy + 3x^2)^2 = \frac{9}{2}x^5$$

$$\Rightarrow 2(y^2 + 3xy + 3x^2)^2 = 9x^5$$

is required solution.

Solve:

❖ **Question # 15:**

$$\frac{dy}{dx} = \frac{x + 3y - 5}{x - y - 1}$$

Solution:

Given equation is

$$\frac{dy}{dx} = \frac{x + 3y - 5}{x - y - 1} \text{ --- (a)}$$

Put

$$x = X + h \quad \& \quad y = Y + k$$

$$\Rightarrow dx = dX \quad \& \quad dy = dY$$

Thus equation (a) becomes

$$\frac{dY}{dX} = \frac{X + h + 3(Y + k) - 5}{X + h - (Y + k) - 1}$$

$$\Rightarrow \frac{dY}{dX} = \frac{X + h + 3Y + 3k - 5}{X + h - Y - k - 1}$$

$$\Rightarrow \frac{dY}{dX} = \frac{X + 3Y + h + 3k - 5}{X - Y + h - k - 1}$$

$$\text{Put } h + 3k - 5 = 0 \text{ --- (*)}$$

$$\& \quad h - k - 1 = 0 \text{ --- (**)}$$

On solving (*) & (**), we have

$$h = -2 \& \quad k = -1$$

Therefore,

$$1 + 1 = B$$

$$\Rightarrow B = 2$$

To find the value of A, we have to solve the (i).

Therefore,

$$1 - v = Av + A + B$$

Comparing the coefficient of v , we have

$$A = 1$$

Therefore,

$$\frac{1 - v}{(v + 1)^2} = \frac{-1}{v + 1} + \frac{2}{(v + 1)^2}$$

$$\text{Now (i)} \Rightarrow \frac{dY}{dX} = \frac{X + 3Y}{X - Y} \quad \dots (b)$$

This is a homogenous differential equation in X & Y . to solve this, put

$$Y = vX$$

$$\Rightarrow \frac{dY}{dX} = v + X \frac{dv}{dX}$$

Thus equation (b) becomes

$$v + X \frac{dv}{dX} = \frac{X + 3vX}{X - vX}$$

$$\Rightarrow v + X \frac{dv}{dX} = \frac{1 + 3v}{1 - v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{1 + 3v}{1 - v} - v$$

$$\Rightarrow X \frac{dv}{dX} = \frac{1 + 3v - v + v^2}{1 - v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{v^2 + 2v + 1}{1 - v}$$

$$\Rightarrow \frac{(1 - v)}{v^2 + 2v + 1} dv = \frac{dX}{X}$$

$$\Rightarrow \frac{(1 - v)}{(v + 1)^2} dv = \frac{dX}{X}$$

Integrating both sides, we have

$$\int \frac{(1 - v)}{(v + 1)^2} dv = \int \frac{dX}{X} \quad \dots (i)$$

Suppose that

$$\frac{(1 - v)}{(v + 1)^2} = \frac{A}{v + 1} + \frac{B}{(v + 1)^2}$$

$$1 - v = A(v + 1) + B \quad \dots (ii)$$

$$\text{Put } v + 1 = 0 \Rightarrow v = -1 \text{ in (ii)}$$

$$\int \left[\frac{-1}{v + 1} + \frac{2}{(v + 1)^2} \right] dv = \int \frac{dX}{X}$$

$$- \int \frac{dv}{v + 1} + 2 \int \frac{dv}{(v + 1)^2} = \int \frac{dX}{X}$$

$$- \ln(v + 1) + 2 \left(\frac{-1}{v + 1} \right) = \ln X + \ln C$$

$$\Rightarrow - \ln \left(\frac{Y}{X} + 1 \right) - \frac{2}{\frac{Y}{X} + 1} = \ln X + \ln C$$

$$\Rightarrow - \ln \left(\frac{X + Y}{X} \right) - \frac{2X}{X + Y} = \ln X + \ln C$$

Putting the values of X and Y , we have

$$- \ln \left(\frac{x - 2 + y - 1}{x - 2} \right) - \frac{2(x - 2)}{x - 2 + y - 1} = \ln(x - 2) + \ln C$$

$$\Rightarrow - \ln \left(\frac{x + y - 3}{x - 2} \right) - \frac{2(x - 2)}{x + y - 3} = \ln(x - 2) + \ln C$$

$$\Rightarrow - \ln(x + y - 3) + \ln(x - 2) - \frac{2(x - 2)}{x + y - 3} = \ln(x - 2) + \ln C$$

$$\Rightarrow -\ln(x+y-3) - \frac{2(x-2)}{x+y-3} = \ln C$$

$$\Rightarrow -\frac{2(x-2)}{x+y-3} = \ln C + \ln(x+y-3)$$

$$\Rightarrow -\frac{2(x-2)}{x+y-3} = \ln C (x+y-3)$$

is required solution.

❖ **Question # 16:**

$$\frac{dy}{dx} = -\frac{4x+3y+15}{2x+y+7}$$

Solution:

Given equation is

$$\frac{dy}{dx} = -\frac{4x+3y+15}{2x+y+7} \quad \text{--- (a)}$$

Put

$$x = X + h \quad \& \quad y = Y + k$$

$$\Rightarrow dx = dX \quad \& \quad dy = dY$$

Thus equation (a) becomes

$$\frac{dY}{dX} = -\left(\frac{4(X+h)+3(Y+k)+15}{2(X+h)+Y+k+7}\right)$$

$$\Rightarrow \frac{dY}{dX} = -\left(\frac{4X+4h+3Y+3k+15}{2X+2h+Y+k+7}\right)$$

$$\Rightarrow \frac{dY}{dX} = \frac{4X+3Y+4h+3k+15}{2X+Y+2h+k+7}$$

$$\text{Put } 4h+3k+15=0 \quad \text{--- (*)}$$

$$\& \quad 2h+k+7=0 \quad \text{--- (**)}$$

On solving (*) & (**), we have

$$h = -3 \& \quad k = -1$$

Therefore,

$$\frac{dY}{dX} = \frac{4X+3Y}{2X+Y} \quad \text{--- (b)}$$

This is a homogenous differential equation in X & Y . to solve this, put

$$Y = vX$$

$$\Rightarrow \frac{dY}{dX} = v + X \frac{dv}{dX}$$

Thus equation (b) becomes

$$v + X \frac{dv}{dX} = -\left(\frac{4X+3vX}{2X+vX}\right)$$

$$\Rightarrow v + X \frac{dv}{dX} = -\left(\frac{4+3v}{2+v}\right)$$

$$\Rightarrow X \frac{dv}{dX} = -\left(\frac{4+3v}{2+v}\right) - v$$

$$\Rightarrow X \frac{dv}{dX} = \frac{-4-3v-2v-v^2}{2+v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{-v^2-5v-4}{2+v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{-(v^2+5v+4)}{2+v}$$

$$\Rightarrow X \frac{dv}{dX} = -\frac{(v+1)(v+4)}{2+v}$$

$$\Rightarrow \frac{2+v}{(v+1)(v+4)} dv = -\frac{dX}{X}$$

Integrating both sides, we have

$$\int \frac{2+v}{(v+1)(v+4)} dv = -\int \frac{dX}{X}$$

Consider,

$$\frac{v+2}{(v+1)(v+4)} = \frac{A}{v+1} + \frac{B}{v+4}$$

$$\Rightarrow v+2 = A(v+4) + B(v+1) \quad \text{--- (i)}$$

$$\text{Put } v+1=0 \Rightarrow v=-1 \text{ in (i)}$$

$$-1 + 2 = A(-1 + 4)$$

$$\Rightarrow A = \frac{1}{3}$$

$$\text{Put } v + 4 = 0 \Rightarrow v = -4 \text{ in (i)}$$

$$-4 + 2 = B(-4 + 1)$$

$$\Rightarrow B = \frac{2}{3}$$

Thus,

$$\frac{v+2}{(v+1)(v+4)} = \frac{1}{3(v+1)} + \frac{2}{3(v+4)}$$

Putting values in (3)

$$\int \left[\frac{1}{3(v+1)} + \frac{2}{3(v+4)} \right] dv = - \int \frac{dX}{X}$$

$$\Rightarrow \frac{1}{3} \int \frac{dv}{v+1} + \frac{2}{3} \int \frac{dv}{v+4} = - \int \frac{dX}{X}$$

$$\Rightarrow \frac{1}{3} \ln(v+1) + \frac{2}{3} \ln(v+4) = -\ln X + \ln C$$

$$\Rightarrow \ln(v+1)^{\frac{1}{3}} + \ln(v+4)^{\frac{2}{3}} = \ln \frac{C}{X}$$

$$\Rightarrow \ln(v+1)^{\frac{1}{3}}(v+4)^{\frac{2}{3}} = \ln \frac{C}{X}$$

$$\Rightarrow (v+1)^{\frac{1}{3}}(v+4)^{\frac{2}{3}} = \frac{C}{X}$$

$$\Rightarrow \left(\frac{Y}{X} + 1 \right)^{\frac{1}{3}} \left(\frac{Y}{X} + 4 \right)^{\frac{2}{3}} = \frac{C}{X}$$

$$\Rightarrow \left(\frac{Y+X}{X} \right)^{\frac{1}{3}} \left(\frac{Y+4X}{X} \right)^{\frac{2}{3}} = \frac{C}{X}$$

$$\Rightarrow (Y+X)^{\frac{1}{3}}(Y+4X)^{\frac{2}{3}} = C$$

Putting the values of X and Y

$$(y+1+x+3)^{\frac{1}{3}}(y+1+4x+12)^{\frac{2}{3}} = c$$

$$\Rightarrow (x+y+4)^{\frac{1}{3}}(y+4x+13)^{\frac{2}{3}} = c$$

$$\Rightarrow (x+y+4)(y+4x+13)^2 = c$$

is required solution.

❖ Question # 17:

$$(3y - 7x - 3)dx + (7y - 3x - 7)dy = 0$$

Solution:

Given equation is

$$(3y - 7x - 3)dx + (7y - 3x - 7)dy = 0$$

$$\frac{dy}{dx} = -\frac{(3y - 7x - 3)}{(7y - 3x - 7)} \quad \text{--- (a)}$$

Put

$$x = X + h \quad \& \quad y = Y + k$$

$$\Rightarrow dx = dX \quad \& \quad dy = dY$$

Thus equation (a) becomes

$$\frac{dY}{dX} = -\frac{3(Y+k) - 7(X+h) - 3}{7(Y+k) - 3(X+h) - 7}$$

$$\Rightarrow \frac{dY}{dX} = -\frac{3Y + 3k - 7X - 7h - 3}{7Y + 7k - 3X - 3h - 7}$$

$$\Rightarrow \frac{dY}{dX} = -\frac{3Y - 7X + 3k - 7h - 3}{7Y - 3X + 7k - 3h - 7}$$

$$\text{Put } 3k - 7h - 3 = 0 \quad \text{--- (*)}$$

$$\& \quad 7k - 3h - 7 = 0 \quad \text{--- (**)}$$

On solving (*) & (**), we have

$$h = 0 \& \quad k = 1$$

Therefore,

$$\frac{dY}{dX} = \frac{4X + 3Y}{2X + Y} \quad \text{--- (b)}$$

This is a homogenous differential equation in X & Y. to solve this, put

$$Y = vX$$

$$\Rightarrow \frac{dY}{dX} = v + X \frac{dv}{dX}$$

Thus equation (b) becomes

$$v + X \frac{dv}{dX} = \frac{7X - 3vX}{-3X + 7vX}$$

$$\Rightarrow v + X \frac{dv}{dX} = \frac{7 - 3v}{-3 + 7v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{7 - 3v}{-3 + 7v} - v$$

$$\Rightarrow X \frac{dv}{dX} = \frac{7 - 3v + 3v - 7v^2}{-3 + 7v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{7(1 - v^2)}{-3 + 7v}$$

$$\Rightarrow \frac{(-3 + 7v)}{7(1 - v^2)} dv = \frac{dX}{X}$$

$$\Rightarrow \frac{7\left(\frac{-3}{7} + v\right)}{7(1 - v^2)} dv = \frac{dX}{X} \dots (c)$$

Consider,

$$\Rightarrow \frac{v - \frac{3}{7}}{1 - v^2} = \frac{A}{1 - v} + \frac{B}{1 + v}$$

$$\Rightarrow v - \frac{3}{7} = A(1 + v) + B(1 - v) \dots (i)$$

Put $1 - V = 0 \Rightarrow V = 1$ in (i), we have

$$1 - \frac{3}{7} = A(2)$$

$$\Rightarrow \frac{4}{7} = A(2)$$

$$\Rightarrow A = \frac{2}{7}$$

Put $1 + V = 0 \Rightarrow V = -1$ in (i)

$$\Rightarrow -1 - \frac{3}{7} = B(2)$$

$$\Rightarrow B = \frac{-5}{7}$$

So,

$$\frac{v - \frac{3}{7}}{1 - v^2} = \frac{2}{7(1 - v)} + \frac{-5}{7(1 + v)}$$

Equation (c) will be

$$\left[\frac{2}{7(1 - v)} - \frac{5}{7(1 + v)} \right] dv = \frac{dX}{X}$$

Integrating both sides, we have

$$\frac{2}{7} \int \frac{dv}{1 - v} - \frac{5}{7} \int \frac{dv}{1 + v} = \int \frac{dX}{X}$$

$$\frac{-2}{7} \ln(1 - v) - \frac{5}{7} \ln(1 + v) = \ln X + \ln C$$

$$(1 - v)^{\frac{-2}{7}} (1 + v)^{\frac{-5}{7}} = CX$$

$$\Rightarrow \left(1 - \frac{Y}{X}\right)^{\frac{-2}{7}} \left(1 + \frac{Y}{X}\right)^{\frac{-5}{7}} = CX \quad \because v = \frac{Y}{X}$$

$$\left(\frac{X - Y}{X}\right)^{\frac{-2}{7}} \left(\frac{X + Y}{X}\right)^{\frac{-5}{7}} = \frac{1}{CX}$$

$$\left(\frac{1}{C}\right)^7 = (X - Y)^2 (X + Y)^5$$

Putting values of X and Y

$$(x - y + 1)^2 (x + y - 1)^5 = c$$

is required solution.

❖ Question # 18:

$$\frac{dy}{dx} = \frac{3x - 4y - 2}{3x - 4y - 3}$$

Solution:

Given equation is

$$\frac{dy}{dx} = \frac{3x - 4y - 2}{3x - 4y - 3} \quad \dots (a)$$

Put

$$t = 3x - 4y$$

$$\Rightarrow \frac{dt}{dx} = 3 - 4 \frac{dy}{dx}$$

$$\Rightarrow 4 \frac{dy}{dx} = 3 - \frac{dt}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3}{4} - \frac{1}{4} \frac{dt}{dx}$$

Thus equation (a) becomes

$$\frac{3}{4} - \frac{1}{4} \frac{dt}{dx} = \frac{t - 2}{t - 3}$$

$$\Rightarrow \frac{1}{4} \frac{dt}{dx} = \frac{3}{4} - \frac{t - 2}{t - 3}$$

$$\Rightarrow \frac{1}{4} \frac{dt}{dx} = \frac{3(t - 3) - 4(t - 2)}{4(t - 3)}$$

$$\Rightarrow \frac{1}{4} \frac{dt}{dx} = \frac{3t - 9 - 4t + 8}{4(t - 3)}$$

$$\Rightarrow \frac{dt}{dx} = \frac{-t - 1}{t - 3}$$

$$\Rightarrow \frac{dt}{dx} = \frac{-(t + 1)}{t - 3}$$

$$\Rightarrow \frac{t - 3}{t + 1} dt = dx$$

Integrating both sides, we have

$$\int \frac{t - 3}{t + 1} dt = \int dx$$

$$\Rightarrow \int \frac{t + 1 - 1 - 3}{t + 1} dt = \int dx$$

$$\Rightarrow \int \frac{t + 1}{t + 1} dt - 4 \int \frac{dt}{t + 1} = \int dx$$

$$\Rightarrow t - 4 \ln(t + 1) = x + c$$

$$\Rightarrow 3x - 4y - 4 \ln(3x - 4y + 1) = x + c$$

is required solution.

❖ Question # 19:

$$\frac{dy}{dx} = \frac{y - x + 1}{y - x + 5}$$

Solution:

Given equation is

$$\frac{dy}{dx} = \frac{y - x + 1}{y - x + 5} \quad \dots (a)$$

Put

$$y - x = t$$

$$\Rightarrow \frac{dy}{dx} - 1 = \frac{dt}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dt}{dx} + 1$$

Thus equation (a) becomes

$$\Rightarrow \frac{dt}{dx} + 1 = \frac{t + 1}{t + 5}$$

$$\Rightarrow \frac{dt}{dx} = \frac{t + 1}{t + 5} - 1$$

$$\Rightarrow \frac{dt}{dx} = \frac{t + 1 - t - 5}{t + 5}$$

$$\Rightarrow \frac{dt}{dx} = \frac{-4}{t + 5}$$

$$\Rightarrow (t + 5)dt = -4dx$$

Integrating both sides, we have

$$\Rightarrow \int (t+5)dt = -4 \int dx$$

$$\Rightarrow \frac{t^2}{2} + 5t = -4x + c$$

$$\Rightarrow \frac{1}{2}(y-x)^2 + 5(y-x) = -4x + c \quad \because t = y-x$$

$$\Rightarrow \frac{1}{2}(y-x)^2 + 5y - 5x = -4x + c$$

$$\Rightarrow \frac{1}{2}(y-x)^2 + 5y - x = c$$

$$\Rightarrow (y-x)^2 - 2x + 10y = 2c$$

$$\Rightarrow (y-x)^2 - 2x + 10y = c'$$

is required solution.

❖ Question # 20:

$$\frac{dy}{dx} = \frac{x-2y+5}{2x+y-1}$$

Solution:

Given equation is

$$\frac{dy}{dx} = \frac{x-2y+5}{2x+y-1} \quad \text{--- (i)}$$

Put

$$x = X + h \quad \& \quad y = Y + k$$

$$\Rightarrow dx = dX \quad \& \quad dy = dY$$

Thus equation (a) becomes

$$\frac{dY}{dX} = \frac{(X+h)-2(Y+k)+5}{2(X+h)+(Y+k)-1}$$

$$\Rightarrow \frac{dY}{dX} = \frac{X+h-2Y-2k+5}{2X+2h+Y+k-1}$$

$$\Rightarrow \frac{dY}{dX} = \frac{X-2Y+h-2k+5}{2X+Y+2h+k-1}$$

$$\text{Put } h-2k+5=0 \quad \text{--- (*)}$$

$$\& \quad 2h+k-1=0 \quad \text{--- (**)}$$

On solving (*) & (**), we have

$$h = -\frac{3}{5} \quad \& \quad k = +\frac{11}{5}$$

Therefore,

$$\frac{dY}{dX} = \frac{X-2Y}{2X+Y} \quad \text{--- (b)}$$

This is a homogenous differential equation in X & Y. to solve this, put

$$Y = vX$$

$$\Rightarrow \frac{dY}{dX} = v + X \frac{dv}{dX}$$

Thus equation (b) becomes

$$v + X \frac{dv}{dX} = \frac{X-2vX}{2X+vX}$$

$$\Rightarrow v + X \frac{dv}{dX} = \frac{1-2v}{2+v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{1-2v}{2+v} - v$$

$$\Rightarrow X \frac{dv}{dX} = \frac{1-2v-2v-v^2}{2+v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{-v^2-4v+1}{2+v}$$

$$\Rightarrow X \frac{dv}{dX} = \frac{-(v^2+4v-1)}{2+v}$$

$$\Rightarrow \frac{v+2}{v^2+4v-1} dv = -\frac{dX}{X}$$

Integrating both sides, we have

$$\int \frac{v+2}{v^2+4v-1} dv = -\int \frac{dX}{X}$$

$$\frac{1}{2} \int \frac{2v + 4}{v^2 + 4v - 1} dv = - \int \frac{dX}{X}$$

$$\frac{1}{2} \ln(v^2 + 4v - 1) = -\ln X + \ln C$$

$$(v^2 + 4v - 1)^{\frac{1}{2}} = \frac{C}{X}$$

$$\sqrt{(v^2 + 4v - 1)} = \frac{C}{X}$$

$$\sqrt{\frac{Y^2}{X^2} + 4\frac{Y}{X} - 1} = \frac{C}{X} \quad \therefore v = \frac{Y}{X}$$

$$\frac{\sqrt{Y^2 + 4XY - X^2}}{X} = \frac{C}{X}$$

$$\sqrt{Y^2 + 4XY - X^2} = C$$

Putting the values of Y and X

$$\sqrt{\left(y - \frac{11}{5}\right)^2 + 4\left(x + \frac{3}{5}\right)\left(y - \frac{11}{5}\right) - \left(x + \frac{3}{5}\right)^2} = C$$

On solving this by simple algebraic formulae and operation, we will finally obtain the answer as follow:

$$x^2 - y^2 - 4xy + 10x + 2y = c$$

is required solution.